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Dense Passage Indexing and Semantic Graph Traversal for Scholarly Literature Discovery

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ABSTRACT

Finding obscure prior art in scholarly literature requires combining dense semantic similarity with multi-hop citation graph navigation. We propose Graph-DPR, a hybrid retrieval engine that unifies dense passage vector indexing with breadth-first citation graph expansion. The system retrieves initial candidate seed papers via HNSW vector indexing and iteratively expands the candidate pool along citation and co-authorship edges. In literature search benchmarks across 2 million academic papers, Graph-DPR improves recall@20 by 22.8% over isolated dense vector retrieval. The main drawback is indexing latency and graph memory footprints when processing densely connected citation hub nodes.

Keywords— Natural Language Processing, Information Retrieval, Dense Retrieval, Graph Traversal

INTRODUCTION

Scholarly research search cannot rely on isolated keyword or embedding matches alone; relevant foundational methods often share conceptual lineage rather than superficial lexical similarity.

METHODOLOGY

Graph-DPR maintains an in-memory FAISS vector index paired with an adjacency graph cache. Initial vector top-k hits seed a personalized PageRank traversal that re-ranks candidate documents based on graph connectivity.

RESULTS

Evaluated on finding missing citations in submitted manuscripts, Graph-DPR retrieved 88.4% of relevant references in the top-20 results, significantly beating BM25 (61.2%) and pure DPR (71.9%).

LIMITATIONS

Citation hubs (papers with over 10,000 citations) create graph explosion during traversal, requiring aggressive neighbor pruning that can occasionally drop subtle niche connections.

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